Prime-field k-sum table against the mitm incumbent, same session and targets

2026-10-29

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Track: generic decomposition (bounded by √r; cannot beat rho). Both arms are generic decomposition oracles (AGENTS.md “Tracks of index calculus”, 2026-10-29). Nothing here moves the algebraic track.

Status: UNQUALIFIED. This is one host and one binary, with no `claims/*.json` record yet. Leaderboard promotion needs a second-host replay (the incumbent study’s PROTOCOL §13) and that record.

Result. The ranking ran on the incumbent’s own four holdout curves, with its 48 targets, its method hashes and its measurement seed. The k = 5 sum table of isogeny-labs/crypto#2852 costs less than the recorded incumbent at every size, at every band weight. The incumbent is the negation-folded mitm pair table of `research/prime_base_oracle_tuning_20261029`.

bits	`ksum5 / inc`, w = 0		w = 0.1		w = 1		registered prediction
20	0.701	[0.661, 0.744]	0.665	[0.624, 0.709]	0.521	[0.480, 0.567]	≈ 0.7
24	0.453	[0.433, 0.475]	0.455	[0.433, 0.480]	0.466	[0.434, 0.501]	≈ 0.45
28	0.331	[0.320, 0.344]	0.327	[0.314, 0.340]	0.305	[0.288, 0.325]	≈ 0.33
32	0.218	[0.211, 0.226]	0.219	[0.211, 0.227]	0.222	[0.211, 0.233]	≈ 0.22

The curves are `icv1-fp20-tm565-6d7d1d32`, `icv1-fp24-tm4739-bb3c4b72`, `icv1-fp28-tm1499-56171ddc` and `icv1-fp32-tm36751-2eaac89e`. The table shows paired ratios of total cost with cluster-bootstrap 95% intervals: 96 pairs per cell, 48 workloads, 4,000 resamples, seed 20261002. The registered primary test (24, 28, 32 bits, `ksum5 < inc` at all three weights) **holds in all three cells**.

What this is.

- **A ranking.** The k-sum method itself landed in #2808 and #2852. This study adds the same-instance measurement that makes it rankable against the incumbent, as the tracks rule requires.
- **What it shows.** On these four curves, the generic track's best recipe is now a k-sum table, not the mitm pair table. The generic-track baseline on them falls by 1.4× (20 bits) to 4.6× (32 bits).
- **What it is compared against.** The recorded incumbent recipe, not a newly tuned pair table. At 32 bits the incumbent's base sat at N = 1024 in both its `inc` and `sel` arms; its own README calls that selection null by construction. The widening gap is consistent with the two generic optima, 5/9 for k = 5 against about 2/3 for the folded pair table (`[derived]`, #2852 §1). One curve per size gives no exponent claim here.

Distance remaining (same session, w = 0):

- `ksum5` is 5.4×, 6.3×, 7.2× and 7.3× rho-neg at 20, 24, 28 and 32 bits;
- it is 7.3×, 6.6×, 6.7× and 7.4× bsgs-neg;
- it is generic, so its limit is BSGS, and it stays above BSGS everywhere.

Public synthetic curves and targets; no security claim. Registered in `PROTOCOL.md` before any timed run; hashes are in `REGISTRATION.sha256`.

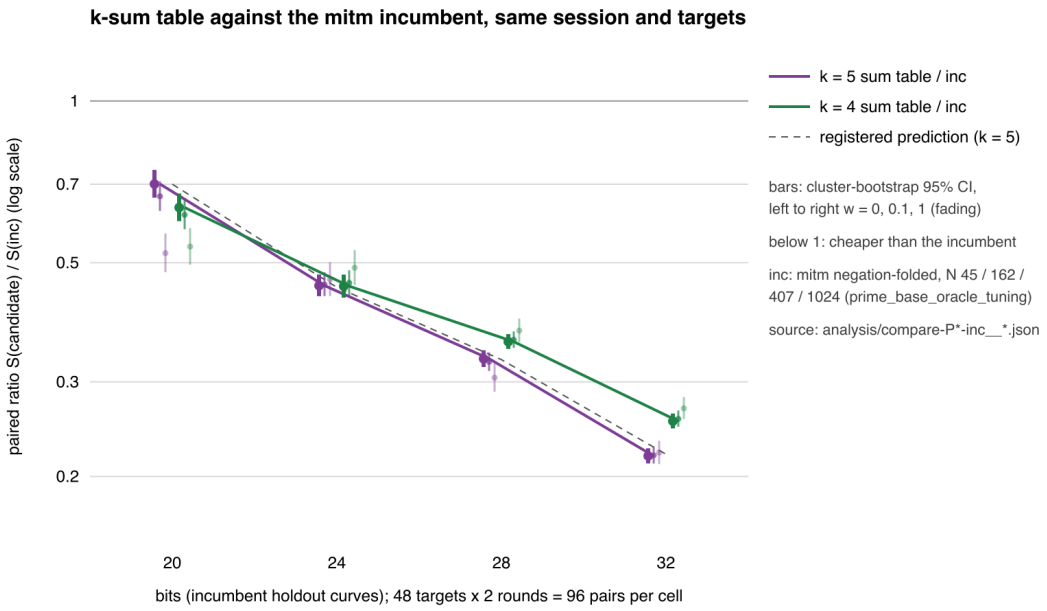


Figure 1: Paired ratio of each k-sum arm to the incumbent, per size, with 95% intervals at w = 0, 0.1 and 1. Dashed: the registered prediction. Source: ratio.svg.

1. Method `[measured]`

- **Instances.** The curves, targets and measurement settings are those of the incumbent's `specs/holdout-P<b>.json`: curve seed 20261018202, target seed 20261018211, mea-

surement seed 20261018221, 2 rounds plus 1 warm-up, alternate order, isolation L0. None was chosen here.

- **Arms.**

arm	role	method
`rho-neg`, `rho-neg-aa`	reference, A/A	`rho.negation`
`bsgs-neg`	reference	`bsgs.negation`
`inc`	baseline	the incumbent recipe, N = 45 / 162 / 407 / 1024; method hashes identical to the incumbent session's
`ksum4`	candidate	A = 10 / 14 / 21 / 30, frozen from #2808′s train curves
`ksum5`	candidate	A = 9 / 11 / 15 / 19, frozen from #2852′s train curves

No parameter was tuned on these curves.

- **Binary.** `ecbench-v2` (`BINARY.sha256`), the #2852 test-panel binary. One binary, one session per size.
- **Accounting parity.** Every arm is one target, cold. The pair table and the k-sum table are both built inside each run and charged by the same counters, so no build is amortised.

2. Integrity `[measured]`

- **Every answer verified, and every replay reproduced.** 4 sessions × 864 records = 3,456 runs, every answer verified. `ecbench verify --replay-all` reproduces all 2,304 deterministic replays with 0 problems (`audits/`).
- **Replication gate (PROTOCOL §4): passed.**
  - `inc` and `rho-neg` reproduce the incumbent's recorded means in its `analysis/band-holdout.md`.
  - The match is exact to 3 decimals: all 24 values, 4 sizes × 2 arms × 3 weights.
  - For example, `inc` at 32 bits gives 32.240 / 35.309 / 62.931.
- **A/A.** `rho-neg` vs `rho-neg-aa` is unresolved inside the band at every size, as it should be.

3. Results `[measured]`

Mean `S` (group-operation equivalents / √r) at w = 0 / 0.1 / 1; full table in `analysis/band.md`.

bits	inc		ksum4		ksum5		rho-neg		bsgs-neg	
20	10.515	/	6.660	/	7.367	/	1.357	/	1.009	/
	11.829	/	7.262	/	7.862	/	1.474	/	1.109	/
	23.655		12.675		12.323		2.529		2.006	
24	13.976	/	6.319	/	6.334	/	1.013	/	0.958	/
	15.168	/	6.954	/	6.906	/	1.104	/	1.054	/
	25.892		12.670		12.054		1.919		1.913	
28	20.478	/	7.298	/	6.785	/	0.939	/	1.015	/
	22.204	/	7.978	/	7.259	/	1.031	/	1.116	/
	37.737		14.101		11.526		1.851		2.028	
32	32.240	/	8.171	/	7.040	/	0.968	/	0.950	/
	35.309	/	9.039	/	7.730	/	1.064	/	1.045	/
	62.931		16.853		13.941		1.930		1.900	

Paired ratios for every registered pair: `analysis/ratios.md`, from `analysis/compare-*.json`. Verdicts from `ecbench band --ranks`:

- **`ksum5 < inc` and `ksum4 < inc` at every weight, at all four sizes.** The primary test, plus the 20-bit and `ksum4` secondaries.
- **`ksum5` vs `ksum4`.** `ksum5 < ksum4` at 28 and 32 bits (0.93 and 0.86 at w = 0). Unresolved at 20 and 24 bits. At 20 bits `ksum4` has the lower point estimate at w = 0 (1.106 [1.057, 1.161]) and w = 0.1, while at w = 1 the interval spans 1. So there is no single best k below 28 bits.
- **`rho-neg < ksum5` and `bsgs-neg < ksum5` everywhere.**

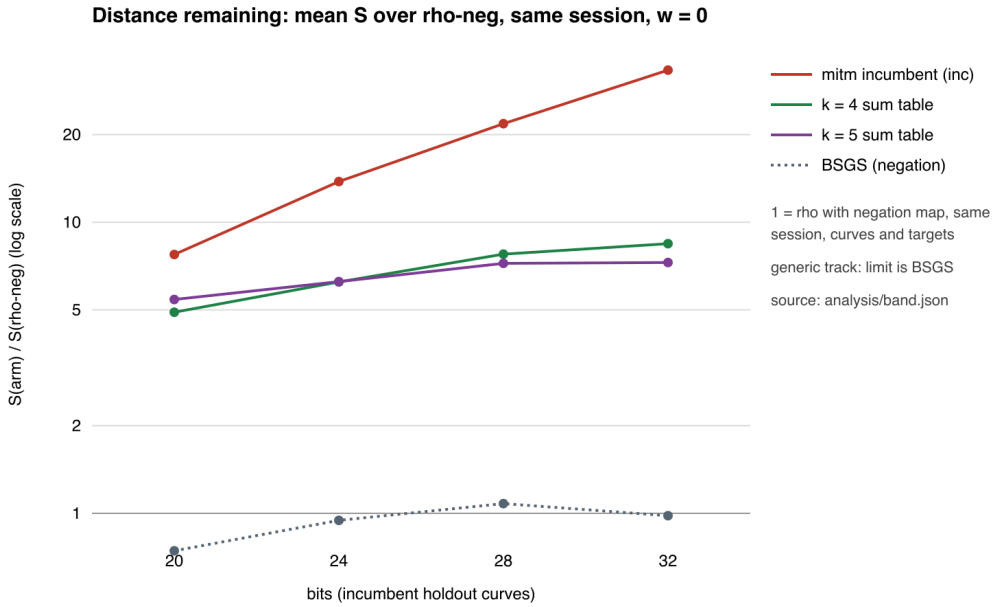


Figure 2: Each arm's mean S over rho-neg in the same session, w = 0. Source: rho_multiple.svg.

4. Reading

1. **Improvement of the generic-track baseline on these curves (registered test).**
   - At 24, 28 and 32 bits the k-sum table costs 0.45×, 0.33× and 0.22× the recorded incumbent, across the band.
   - The 20-bit cell is secondary. It holds at 0.70× (0.52× at w = 1), but the lower-cost k there is unresolved.
   - Every registered prediction lies inside its interval.
2. **The 20-bit cell is the fragile one.** Open isogeny-labs/crypto#2826 (stride-renew relation targets) reports a 16% cut to the incumbent at 20 bits. If it merges, the 20-bit ratio would move to about 0.84, still below 1. That would need a re-check against the post-#2826 recipe. At 24–32 bits, the other lane's open constant gains (6–22%) cannot close a 2.2–4.6× gap.
3. **This does not touch the algebraic track or rho.** Both arms are generic, and the best of them is still 5.4–7.3× rho on these curves.
4. **Wall time: recorded, not admitted** (L0, shared host at load 10–17).

bits	end-to-end wall `ksum5 / inc`	ns per GAE `ksum5 / inc`
20	0.482 [0.411, 0.566]	0.689 [0.600, 0.790]
24	0.474 [0.424, 0.530]	1.046 [0.953, 1.148]
28	0.366 [0.342, 0.392]	1.106 [1.053, 1.160]
32	0.226 [0.215, 0.237]	1.035 [1.010, 1.060]

The wall ratios follow the counted ratios.

5. Limits

- **Coverage.** One curve per size and one host. No exponent and no pooled claim.
- **Unit.** Counted unit only: ecbench GAE with §5a pricing of native work. The k-sum lookups are hash probes priced inside that band.
- **Baseline.** The incumbent is the recorded recipe. A pair table retuned at 32 bits would narrow that cell's gap; the derived exponents say it would not close it.
- **Status.** UNQUALIFIED: same host. The second-host replay and a `claims/*.json` record are owed before any leaderboard change.

6. Next checks

- **Second-host replay.** Replay the four sessions on isolab-eks or the independent runner (the incumbent's PROTOCOL §13 route), then file `claims/*.json`.
- **Re-check 20 bits** against the stride-renew recipe if #2826 merges.
- **A fresh holdout curve per size,** with A frozen as here, so the result does not rest on one curve.

7. Files

- `PROTOCOL.md`, `REGISTRATION.sha256`, `registered-at.txt`: the registration.
- `specs/P<b>.json`: the four frozen specs.
- `run.sh`: runs the four sessions and their `verify --replay-all` audits. `analyse.sh` produces the band, comparisons, ratio table and figures. Both are thin `ecbench` orchestration; `ECBENCH=` selects the binary.
- `sessions/P<b>/`: native session records. The repository bot may move them to the artifact store.
- `audits/P<b>.txt`: the `verify --replay-all` receipts.
- `analysis/band.md`, `analysis/band.json`: `ecbench band --weights 0,0.1,1 --ranks`.
- `analysis/compare-P<b>-<a>__<b>-w<w>.json`, `analysis/ratios.md`: the paired comparisons.
- `plot.mjs`, `ratio.svg/png`, `rho_multiple.svg/png`: the figures (tooling).
- `dashboard.mjs`: the §7a update (tooling).

  - **Progress-chart point.** One point for the session's best cold one-target IC cell: `ksum4` at 20 bits, 4.91× rho-neg. Its "was" is 7.75×, the incumbent on the same curve and targets.
  - **Other panels.** It adds the plotted-point row and a panel paragraph. The best-of row is unchanged: it keeps #2852′s 40-bit frontier-comparable cell, and no cell here is at 40 bits.
  - **Leaderboard.** No change until the second-host replay.
- `report.pdf`: this README.